

Q1: Find the periodic of the function:

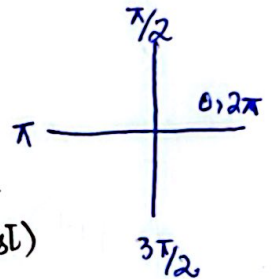
$$f(t) = \cos t$$

Sol: From the Trigonometric formula:

$$\cos(\theta + 2m\pi) = \cos \theta \cos 2m\pi - \sin \theta \sin 2m\pi$$

$$\text{Since } \cos 2m\pi = 1$$

$$\text{And } \sin 2m\pi = 0 \text{ for any integer } m_1 \text{ (smallest)}$$



$$\text{So: } \cos(\theta + 2m\pi) = \cos \theta$$

↳ $\cos \theta$ is a periodic function with period 2π .

Q2: Find period of the function $f(t) = \cos 2t$.

$$\text{Sol: } \cos 2(t+T) = \cos 2t$$

$$\cos(2t + 2T) = \cos 2t$$

Hence:

The period of the function $f(t) = \cos 2t$ is π .

$$\text{Here: } 2T = 2\pi \Rightarrow \boxed{T = \pi}$$

Q3: Find period of $f(t) = \cos t/3 + \cos t/4$

$$\text{Sol: } \cos \frac{1}{3}(t+T) + \cos \frac{1}{4}(t+T) = \cos t/3 + \cos t/4$$

$$\frac{1}{3}T = 2\pi m$$

$$T = 6m\pi$$

$$\frac{1}{4}T = 2n\pi$$

$$T = 8n\pi, n=3$$

$$T = 24\pi$$

→ We have to find the smallest.

$$\text{If } m=4: \boxed{T = 24\pi}$$

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⇒ Properties of periodic functions:

If $f(t+T) = f(t)$, we have

$$\int_a^B f(t) dt = \int_{a+T}^{B+T} f(t) dt$$

For 'a' & 'B' ∈ 'a'

$$\int_0^T f(t) dt = \int_a^{a+T} f(t) dt$$

Q: $\int_{2\pi}^{3\pi} \sin t dt \Rightarrow \text{Sol. } -\cos t \Big|_{2\pi}^{3\pi} \Rightarrow 2 \text{ Ans.}$

⇒ Fourier series: Let the function $f(t)$ be periodic with period 'T'. The function can then be represented by the Trigonometric series.

$$f(t) = \frac{1}{2} a_0 + a_1 \cos \omega_0 t + a_2 \cos 2\omega_0 t + \dots + b_1 \sin \omega_0 t + b_2 \sin 2\omega_0 t + \dots$$

$$= \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos n \omega_0 t + b_n \sin n \omega_0 t) \quad \text{--- (1)}$$

where : $\omega_0 = 2\pi/T$

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$$a_0 = \frac{2}{T} \int_{-T/2}^{T/2} f(t) dt$$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos n\omega_0 t dt$$

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin n\omega_0 t dt$$

A series such as (1) is called Trigonometric Fourier series.

Q. Derive the harmonic form the Trigonometric Fourier series and express the harmonic amplitudes C_n and the phase angle θ in terms of the Fourier coefficients a_n and b_n .

Sol. Consider : $a_n \cos n\omega_0 t + b_n \sin n\omega_0 t$

$$= \sqrt{a_n^2 + b_n^2} \left[\frac{a_n}{\sqrt{a_n^2 + b_n^2}} \cos n\omega_0 t + \frac{b_n}{\sqrt{a_n^2 + b_n^2}} \sin n\omega_0 t \right]$$

$$= C_n [\cos \theta_n \cos n\omega_0 t + \sin \theta_n \sin n\omega_0 t]$$

$$= C_n \cos(n\omega_0 t - \theta_n) \quad \text{Now from eq (1):}$$

$$f(t) = c_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t - \theta_n)$$

$$\because \cos \theta_n = \frac{a_n}{\sqrt{a_n^2 + b_n^2}}$$

$$\because \sin \theta_n = \frac{b_n}{\sqrt{a_n^2 + b_n^2}}$$

$$\because \tan \theta_n = b_n / a_n$$

$$\because \theta_n = \tan^{-1} (b_n / a_n)$$

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Q: Find the Fourier series for the function $f(t)$ defined by:

$$f(t) = \begin{cases} -1 & , \quad -T/2 < t < 0 \\ +1 & , \quad 0 < t < T/2 \end{cases}$$

and $f(t+T) = f(t)$

Sol. $\frac{1}{2} a_0 = \frac{1}{2} \cdot \frac{2}{T} \int_{-T/2}^{T/2} f(t) dt$

$$= \frac{1}{T} \int_{-T/2}^0 f(t) dt + \frac{1}{T} \int_0^{T/2} f(t) dt$$

$$= \frac{1}{T} \int_{-T/2}^0 -dt + \frac{1}{T} \int_0^{T/2} dt \Rightarrow = \frac{1}{T} \left[-t \Big|_{-T/2}^0 + t \Big|_0^{T/2} \right] = 0$$

Now for a_n : $a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos n\omega_0 t dt$

$$a_n = \frac{2}{T} \left[\int_{-T/2}^0 -\cos n\omega_0 t dt + \int_0^{T/2} \cos n\omega_0 t dt \right]$$

$$= \frac{2}{T} \left[-\frac{1}{n\omega_0} \sin n\omega_0 t \Big|_{-T/2}^0 + \frac{1}{n\omega_0} \sin n\omega_0 t \Big|_0^{T/2} \right]$$

$a_n = 0$ Ans.

$$\therefore f(t) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

$$\therefore \omega_0 = 2\pi/T$$

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Now b_n :

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin n\omega_0 t \, dt$$

$$b_n = \frac{2}{T} \left[\int_{-T/2}^0 -\sin n\omega_0 t \, dt + \int_0^{T/2} \sin n\omega_0 t \, dt \right]$$

$$= \frac{2}{T} \left[\frac{+1}{n\omega_0} \cos n\omega_0 t \Big|_{-T/2}^0 + \frac{1}{n\omega_0} \cos n\omega_0 t \, dt \Big|_0^{T/2} \right]$$

$$= \frac{2}{T n \omega_0} \left[1 - \cos(-n\pi) - (\cos n\pi - 1) \right]$$

$$= \frac{2}{T n \omega_0} (1 - \cos n\pi - \cos n\pi + 1)$$

$$\because \cos(-n\pi) = \cos n\pi$$

$$= \frac{2}{T n \omega_0} (2 - 2 \cos n\pi) \Rightarrow = \frac{2}{T n \left(\frac{2\pi}{T}\right)} 2 [1 - \cos n\pi]$$

$$= \frac{2}{n\pi} (1 - \cos n\pi)$$

$$\because \cos n\pi = (-1)^n$$

$$= \begin{cases} \frac{4}{n\pi} & , \quad n \text{ is odd} \\ 0 & , \quad n \text{ is even} \end{cases}$$

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Hence series become:

$$f(t) = \frac{4}{\pi} \sum_{n=\text{odd}}^{\infty} \frac{1}{n} \sin n\omega_0 t \quad \text{Ans.}$$

Q: Find the fourier series for the function $f(t)$ defined by:

$$f(t) = \begin{cases} 0 & -\pi < t < 0 \\ t/\pi & 0 < t < \pi \end{cases}$$

$$\text{and } f(t+2\pi) = f(t)$$

Sol. Given $T = 2\pi$.

$$a_0 = \frac{2}{T} \int_{-T/2}^{T/2} f(t) dt$$

$$a_0 = \frac{2}{2\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$a_0 = \frac{1}{\pi} \left[\underbrace{\int_{-\pi}^0 f(t) dt}_{\rightarrow 0} + \int_0^{\pi} f(t) dt \right]$$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} t/\pi dt$$

$$\begin{aligned} a_0 &= \frac{1}{\pi^2} \int_0^{\pi} t dt \\ &= \frac{1}{\pi^2} \left. \frac{t^2}{2} \right|_0^{\pi} \\ &= \frac{1}{2\pi^2} (\pi^2 - 0) \end{aligned}$$

$$\boxed{a_0 = 1/2}$$

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$$\text{Now } a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt \, dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 0 \cos nt \, dt + \frac{1}{\pi} \int_0^{\pi} \frac{t}{\pi} \cos nt \, dt$$

$$= \frac{1}{\pi^2} \int_0^{\pi} t \cos nt \, dt$$

$$= \frac{1}{\pi^2} \left[t \frac{\sin nt}{n} \Big|_0^{\pi} - \frac{1}{n} \int_0^{\pi} \sin nt \, dt \right]$$

$$= \frac{1}{\pi^2} \left[0 - \frac{1}{n} \frac{\cos nt}{-n} \Big|_0^{\pi} \right]$$

$$= \frac{1}{n^2 \pi^2} [\cos n\pi - \cos 0]$$

$$= \frac{1}{n^2 \pi^2} [\cos n\pi - 1]$$

$$\text{So } a_n = \begin{cases} 0 & \text{even } n \\ -\frac{2}{n^2 \pi^2} & \text{odd } n \end{cases}$$

$$\text{Now } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt \, dt$$

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$$= \frac{1}{\pi} \left[\int_{-\pi}^0 0 \sin nt \, dt + \int_0^{\pi} \sin nt \cdot \frac{1}{\pi} t \, dt \right]$$

$$= \frac{1}{\pi^2} \int_0^{\pi} \sin nt \, dt$$

$$b_n = \frac{1}{\pi^2} \left[t \cdot \left(\frac{-\cos nt}{n} \right) \Big|_0^{\pi} - \left\{ \int_0^{\pi} -\frac{\cos nt}{n} \, dt \right\} \right]$$

$$= \frac{1}{\pi^2} \left[-\frac{\pi}{n} \cos n\pi + \frac{1}{n} \int_0^{\pi} \cos nt \, dt \right]$$

$$= -\frac{1}{n\pi} \cos n\pi + \frac{1}{n\pi^2}$$

$$\Rightarrow \text{Fourier Transform: } F\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{ikx} \, dx$$

Q: Find the fourier series of:

$$f(x) = e^{-x^2}$$

$$\text{Sol: } F\{e^{-x^2}\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2} \cdot e^{ikn} \, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x^2 - ikn)} \, dx$$

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$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x^2 - ikx + \frac{(ik)^2}{2} - \frac{(ik)^2}{2})} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left[\left(x - \frac{ik}{2}\right)^2 - \frac{i^2 k^2}{4}\right]} dx \Rightarrow = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\left[\left(x - \frac{ik}{2}\right)^2 + \frac{k^2}{4}\right]} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(x - \frac{ik}{2}\right)^2} e^{-k^2/4} dx \Rightarrow = \frac{1}{\sqrt{2\pi}} e^{-k^2/4} \int_{-\infty}^{\infty} e^{-\left(x - \frac{ik}{2}\right)^2} dx$$

Put $x - \frac{ik}{2} = u$, $dx = du$

$$= \frac{1}{\sqrt{2\pi}} e^{-k^2/4} \int_{-\infty}^{\infty} e^{-u^2} du$$

$$= \frac{1}{\sqrt{2\pi}} e^{-k^2/4} \sqrt{\pi} \Rightarrow = \frac{1}{\sqrt{2}} e^{-k^2/4} \text{ Ans.}$$

$\Rightarrow$ Fourier Transform:

$\hookrightarrow$ Fourier sine Transform:

$$F_s \{ f(x) \} = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin kx dx$$

$\hookrightarrow$ Fourier cosine Transform:

$$F_c \{ f(x) \} = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos kx dx$$

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Q: Calculate fourier sine transform

$$f(x) = \begin{cases} \sin x & , 0 \leq x \leq \pi \\ 0 & , x > \pi \end{cases}$$

Sol. $F_s \{ f(x) \} = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin kx \, dx$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\pi} (\sin x) \sin kx \, dx + \sqrt{\frac{2}{\pi}} \int_{\pi}^{\infty} (0) \sin kx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\pi} \sin x \sin kx \, dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\pi} \frac{\cos(kx - x) - \cos(kx + x)}{2} \, dx$$

$$= \sqrt{\frac{2}{\pi}} \cdot \frac{1}{2} \int_0^{\pi} \cos(k-1)x - \cos(k+1)x \, dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{\sin(k-1)x}{k-1} - \frac{\sin(k+1)x}{k+1} \right]_0^{\pi}$$

$$= \sqrt{\frac{2}{\pi}} \frac{\sin k\pi}{1-k^2}$$

Ans.

$$\begin{aligned} \because \cos(a+b) &= \cos a \cos b - \sin a \sin b \\ \therefore \cos(a-b) &= \cos a \cos b + \sin a \sin b \end{aligned}$$

$$2\cos(a+b) - \cos(a-b) = -2\sin a \sin b$$

$$\sin a \sin b = \frac{\cos(a+b) - \cos(a-b)}{-2}$$

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Q: Calculate Fourier sine transform:

$$f(x) = \begin{cases} \sin x & , \quad 0 \leq x \leq \pi/2 \\ 0 & , \quad x > \pi/2 \end{cases}$$

Sol. $F_s \{f(x)\} = \sqrt{\frac{2}{\pi}} \int_0^{\pi/2} \sin x \sin kx dx + \int_{\pi/2}^{\infty} (0) \sin kx dx$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\pi/2} \sin x \sin kx dx$$
$$= \sqrt{\frac{2}{\pi}} \left[\frac{\sin(k-1)x}{k-1} - \frac{\sin(k+1)x}{k+1} \right]_0^{\pi/2}$$
$$= \sqrt{\frac{2}{\pi}} \frac{k \cos k \pi/2}{1-k^2}$$

Q: Calculate Fourier cosine transform:

$$f(x) = \begin{cases} \cos x & , \quad 0 \leq x \leq \pi \\ 0 & , \quad x > \pi \end{cases}$$

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Sol. $F_c \{ f(x) \} = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \cos kx dx + \sqrt{\frac{2}{\pi}} \int_{-\infty}^0 f(x) \cos kx dx$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\pi} \cos x \cos kx dx$$

$$\because \cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$$

$$= \sqrt{\frac{2}{\pi}} \cdot \frac{1}{2} \int_0^{\pi} [\cos(k+1)x + \cos(k-1)x] dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{\sin(k+1)x}{k+1} + \frac{\sin(k-1)x}{k-1} \right]_0^{\pi}$$

$$= \sqrt{\frac{2}{\pi}} \frac{k \sin k\pi}{1-k^2}$$

Q: Calculate Fourier cosine Transform:

$$f(x) = \begin{cases} 1 & , 0 \leq x \leq \pi/2 \\ 0 & , x > \pi/2 \end{cases}$$

Sol. $= \sqrt{\frac{2}{\pi}} \int_0^{\pi/2} \cos kx dx$

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$$= \sqrt{\frac{2}{\pi}} \frac{\sin kx}{k} \Big|_0^{\pi/2} \Rightarrow \sqrt{\frac{2}{\pi}} \frac{\sin k \pi/2}{k} \text{ Ans.}$$

Q: Calculate Fourier sine transform:

$$f(x) = \begin{cases} 1 & , 0 \leq x \leq \pi/2 \\ 0 & , x > \pi/2 \end{cases}$$

$$\text{Sol. } F_s \{ f(x) \} = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin kx dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\pi/2} \sin kx dx + \sqrt{\frac{2}{\pi}} \int_{\pi/2}^{\infty} (0) \sin kx dx$$

$$= \sqrt{\frac{2}{\pi}} \left[-\frac{\cos kx}{k} \right]_0^{\pi/2}$$

$$= \sqrt{\frac{2}{\pi}} \left(\frac{1 - \cos \pi/2}{k} \right) \text{ Ans.}$$

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Q: Calculate Fourier sine transform:

$$f(x) = \begin{cases} x, & 0 \leq x \leq \pi/2 \\ 0, & x > \pi/2 \end{cases}$$

$$\text{Sol. } f_s \{f(x)\} = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin kx dx + \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(x) \sin kx dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\pi/2} x (\sin kx) dx + \sqrt{\frac{2}{\pi}} \int_{\pi/2}^{\infty} 0 (\sin kx) dx$$

$$= \sqrt{2/\pi} \int_0^{\pi/2} x \sin kx dx \Rightarrow \text{By parts}$$

$$= \sqrt{\frac{2}{\pi}} \left[x \left(\frac{-\cos kx}{k} \right) \Big|_0^{\pi/2} - \int_0^{\pi/2} \frac{-\cos kx}{k} dx \right]$$

$$= \frac{\pi}{2k} \left[-\cos k \frac{\pi}{2} \right] + \frac{1}{k} \int_0^{\pi/2} \cos kx dx$$

$$= \frac{\pi}{2k} \left[-\cos k \frac{\pi}{2} \right] + \frac{1}{k^2} \left[\sin k \frac{\pi}{2} - \sin k(0) \right]$$

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$$= \sqrt{\frac{2}{\pi}} \left[\frac{\sin k \pi/2}{k^2} - \frac{\pi \cos k \pi/2}{2\pi} \right].$$

Q. Calculate Fourier cosine Transform.

$$f(x) = \begin{cases} x & , 0 \leq x \leq \pi/2 \\ 0 & , x > \pi/2 \end{cases}$$

$$\text{Sol. } \sqrt{\frac{2}{\pi}} \int_0^{\pi/2} x \cos kx dx + \sqrt{\frac{2}{\pi}} \int_{\pi/2}^{\infty} x \cos kx dx$$

$$= x \frac{\sin kx}{k} \Big|_0^{\pi/2} - \int_0^{\pi/2} \frac{\sin kx}{k} dx \Rightarrow = \frac{\pi}{2k} \sin k \frac{\pi}{2}$$

$$= \frac{\pi}{2k} \sin k \frac{\pi}{2} - \frac{1}{k} \left[-\frac{\cos kx}{k} \right]_0^{\pi/2} - \frac{1}{k} \int_0^{\pi/2} \sin kx dx$$

$$= \left[\frac{\pi}{2k} \sin k \frac{\pi}{2} + \frac{1}{k^2} \cos k \frac{\pi}{2} - \frac{1}{k^2} \right] \sqrt{\frac{2}{\pi}}$$

Ans

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